

ECE148: Signal Processing Application

Assignment 4: Fourier Series and Spectrum Estimation

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1 Introduction

Consider a real and periodic signal $f(t)$, of period T . The signal has 7 harmonics, for $m = 1, 2, \dots, 7$.

$$f(t) = \sum_{m=1}^7 a_m \sin\left(\frac{2m\pi t}{T}\right)$$

where the 7 Fourier coefficients $\{a_m\}$ are formed with your personal *7-digit UCSB perm number*. Then we take 32 uniform samples within one period with sample spacing $\Delta t = \frac{T}{32}$ to form a short 32-point sequence $\{f(n)\} = \{f(n\Delta t)\}$, where $n = 0, 1, 2, \dots, 31$. Subsequently, to observe the spectrum of the function $f(t)$, we take a 32-point DFT of the sequence to obtain the 32-point spectral sequence $F(k)$, where $k = 0, 1, 2, \dots, 31$.

$$F(k) = DFT_{n=32}\{f(n)\}$$

2 Fourier Series Expansion

Given:

$$f(t) = \sum_{m=1}^7 a_m \sin\left(\frac{2m\pi t}{T}\right)$$

we can formulate the periodic signal in the form of a complex Fourier series expansion:

$$\begin{aligned} f(t) &= \sum_{m=1}^7 a_m \frac{1}{2j} (e^{j2\pi \frac{mt}{T}} - e^{-j2\pi \frac{mt}{T}}) \\ f(t) &= \sum_{m=1}^7 \frac{a_m}{2j} e^{j2\pi \frac{mt}{T}} - \sum_{m=1}^7 \frac{a_m}{2j} e^{-j2\pi \frac{mt}{T}} \end{aligned}$$

We can rewrite this as:

$$f(t) = \sum_{m=-7}^7 F_m e^{j2\pi \frac{mt}{T}}$$

where the complex Fourier coefficients F_m are:

$$F_m = \begin{cases} \frac{a_m}{2j}, & m > 0 \\ -\frac{a_m}{2j}, & m < 0 \\ 0, & m = 0 \end{cases}$$

Using my perm number, A0V5C88 (modified to 1055388), this is the corresponding plot of the Fourier coefficients.

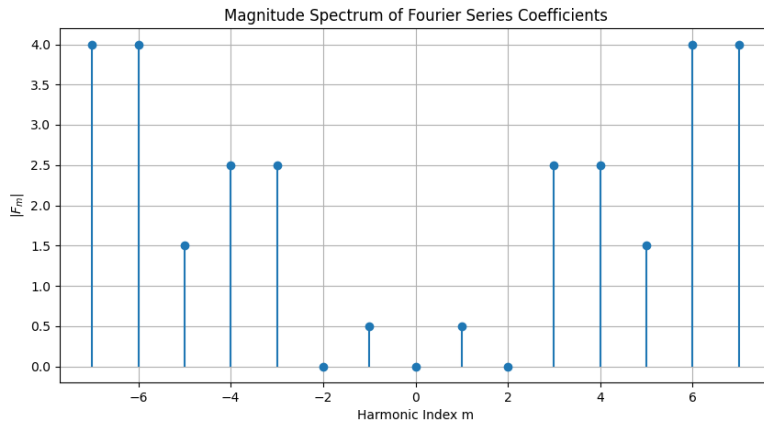


Figure 1: Plot of Fourier Coefficients F_m as a sequence of m

Harmonic m	Fourier Coefficient $ F_m $
± 1	0.5
± 2	0
± 3	2.5
± 4	2.5
± 5	1.5
± 6	4
± 7	4

Table 1: List of Fourier Coefficients F_m

The corresponding physical frequencies ω_m occur at $\omega_m = \frac{2\pi m}{T}$. They are as follows:

Harmonic m	Physical Frequency ω_m
± 1	$\frac{2\pi}{T}$
± 2	$\frac{4\pi}{T}$
± 3	$\frac{6\pi}{T}$
± 4	$\frac{8\pi}{T}$
± 5	$\frac{10\pi}{T}$
± 6	$\frac{12\pi}{T}$
± 7	$\frac{14\pi}{T}$

Table 2: List of Fourier Coefficients F_m

3 Time Function and Sampling

Given:

$$f(t) = \sum_{m=1}^7 a_m \sin\left(\frac{2m\pi t}{T}\right)$$

where a_m is:

$$[1, 0, 5, 5, 3, 8, 8]$$

and $T = 1$, we have the plot of one full period of the time function $f(t)$:

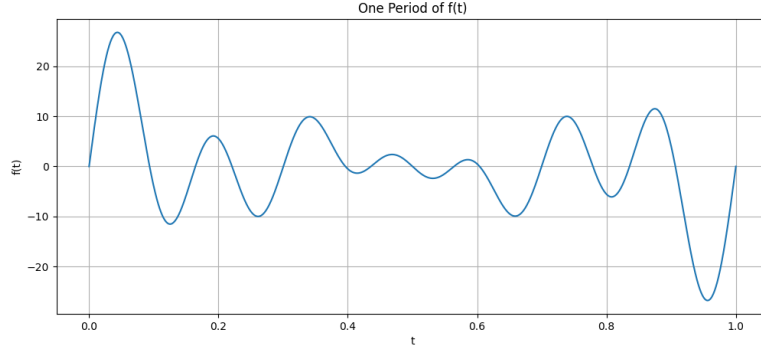


Figure 2: One full period of the time function $f(t)$, from $t = 0$ to $t = T$

Sampling $f(t)$ at $t = n\Delta t$, where $\Delta t = \frac{T}{32}$, we have 32-point sample sequence:

$$f[n] = \sum_{m=1}^7 a_m \sin\left(\frac{2\pi m}{32}n\right)$$

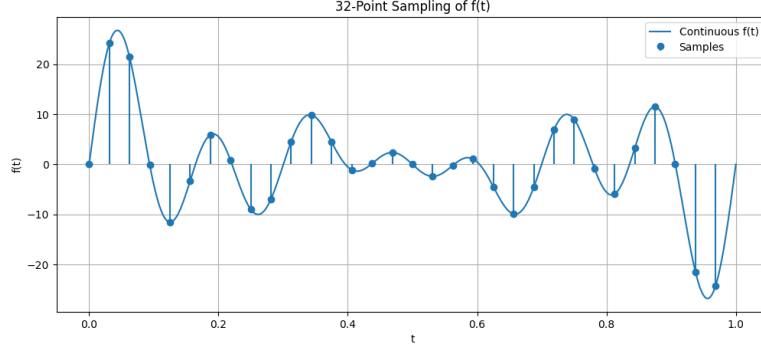


Figure 3: The 32-point sample sequence $f[n]$

4 Fourier Transform of Periodic Functions

Given:

$$f(t) = \sum_{m=1}^7 a_m \sin\left(\frac{2m\pi t}{T}\right) = \sum_{m=1}^7 \frac{a_m}{2j} (e^{j2\pi \frac{m}{T}t} - e^{-j2\pi \frac{m}{T}t})$$

and:

$$\mathcal{F}\{e^{j\omega_0 t}\} = 2\pi\delta(\omega - \omega_0)$$

The Fourier Transform $F(j\omega)$ of the periodic function $f(t)$ is:

$$F(j\omega) = \sum_{m=1}^7 \frac{a_m \pi}{j} [\delta(\omega - m\frac{2\pi}{T}) - \delta(\omega + m\frac{2\pi}{T})]$$

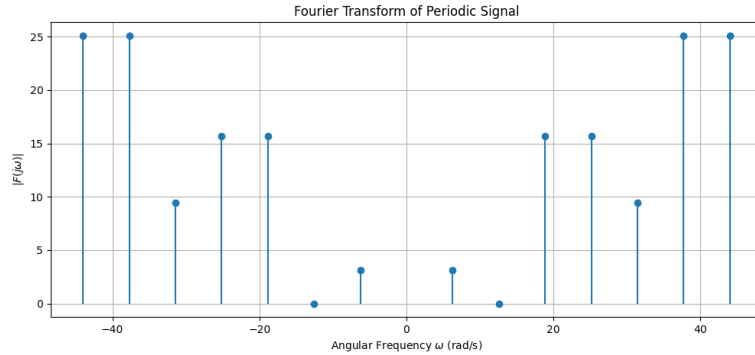


Figure 4: Fourier Transform $F(j\omega)$ of $f(t)$

The spectrum contains impulses at $\omega = \pm m\omega_0$ for $m = 1, 2, 3, 4, 5, 6, 7$.

Harmonic	Peak Location	Amplitude
1	$\pm\omega_0$	π
2	$\pm2\omega_0$	0
3	$\pm3\omega_0$	5π
4	$\pm4\omega_0$	5π
5	$\pm5\omega_0$	3π
6	$\pm6\omega_0$	8π
7	$\pm7\omega_0$	8π

Table 3: Amplitudes and physical frequencies of the peaks

5 DTFT

For the 32-point sampled sequence $f[n] = f(n\Delta t)$ with $\Delta t = \frac{T}{32}$, the discrete-time signal becomes:

$$f[n] = \sum_{m=1}^7 a_m \sin\left(\frac{\pi m}{16} n\right)$$

with peaks occurring at $\Omega_m = \pm m\frac{\pi}{16}$ for $m = 1, 2, 3, 4, 5, 6, 7$.

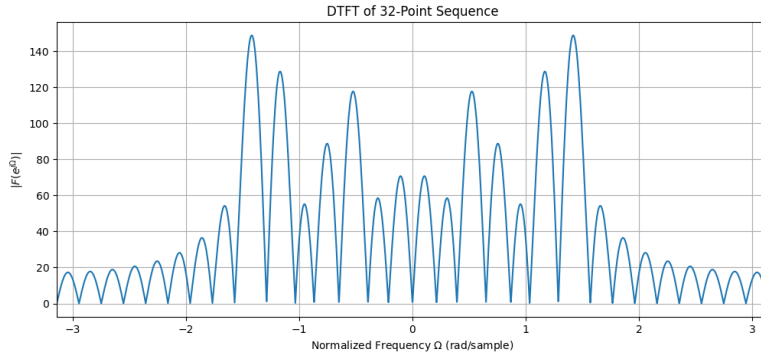


Figure 5: The DTFT $F(e^{j\theta})$ of the 32-point sequence $f[n]$

The peaks will have an amplitude of approximately $16a_m$ because we only have 32 samples.

Harmonic	Peak Location	Amplitude
1	$\pm \frac{1\pi}{16}$	16
2	$\pm \frac{2\pi}{16}$	0
3	$\pm \frac{3\pi}{16}$	80
4	$\pm \frac{4\pi}{16}$	80
5	$\pm \frac{5\pi}{16}$	48
6	$\pm \frac{6\pi}{16}$	128
7	$\pm \frac{7\pi}{16}$	128

Table 4: Amplitude and location of peaks

6 DFT

The 32-point DFT of $f[n]$ is:

$$F(k) = \sum_{n=0}^{31} \sum_{m=1}^7 a_m \sin\left(\frac{2\pi m}{32}n\right) e^{-j\frac{2\pi n}{32}k}$$

$$F(k) = \sum_{n=0}^{31} \sum_{m=1}^7 \frac{a_m}{2j} (e^{j\frac{2\pi m}{32}n} - e^{-j\frac{2\pi m}{32}n}) e^{-j\frac{2\pi n}{32}k}$$

$$F(k) = \sum_{n=0}^{31} \sum_{m=1}^7 \frac{a_m}{2j} (e^{-j\frac{2\pi n}{32}(k-m)} - e^{-j\frac{2\pi n}{32}(k+m)})$$

Peaks will occur at positive frequencies $k = m$ and negative frequencies $k = 32 - m$ for $k = 1, 2, 3, 4, 5, 6, 7$ with amplitude $16a_m$.

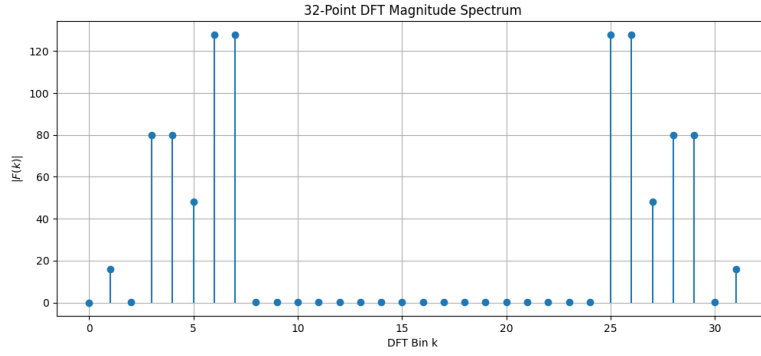


Figure 6: 32-point DFT spectral sequence $F(k)$

Harmonic	k	Amplitude
1	1, 31	16
2	2, 30	0
3	3, 29	80
4	4, 28	80
5	5, 27	48
6	6, 26	128
7	7, 25	128

Table 5: Amplitudes and locations of the peaks

7 Spectral Analysis

Given T , the period of the original function, sampling spacing $\Delta t = \frac{T}{32}$, and DFT length $N = 32$, we can formulate the correspondences to map the integer index k in the DFT spectrum $F(k)$ back to the original physical frequency index ω in $F(j\omega)$. The DTFT frequency variable is frequency $\Omega = \omega\Delta t$ and the DFT samples the DTFT at $\Omega = \frac{2\pi k}{N}$. We substitute and set the equations equal to each other:

$$\omega \frac{T}{32} = \frac{2\pi k}{32}$$

Therefore:

$$\omega = \frac{2\pi k}{T}$$

In the time domain, frequencies occurred at:

$$\omega = \frac{2\pi m}{T}$$

Thus, we have $k = m$. This means that the DFT bin index directly corresponds to the harmonic number of the continuous-time signal. To recover the original harmonic amplitude a_m , divide the DFT peak magnitude by 16:

$$a_m = \frac{|F(k)|}{16}$$